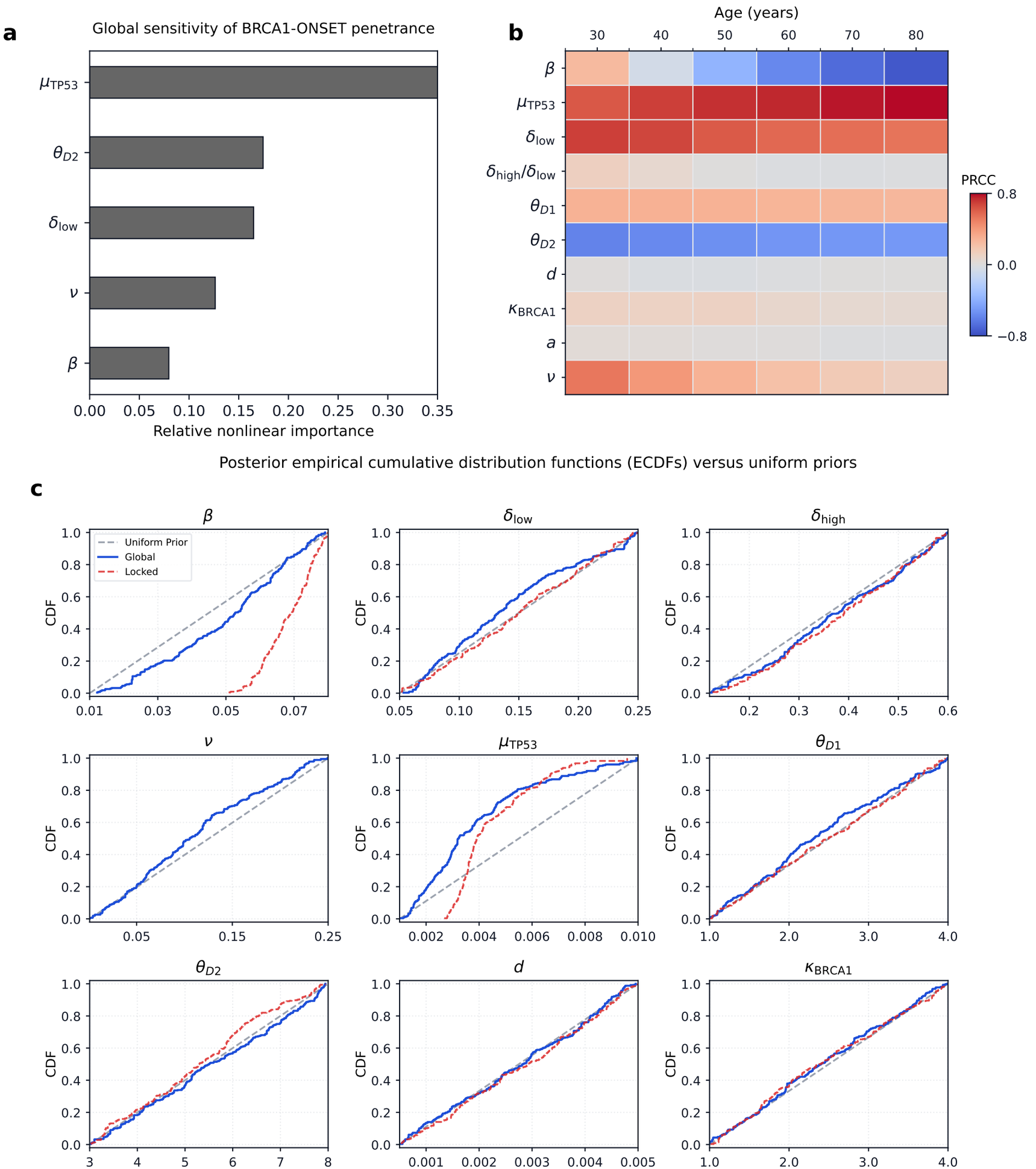
**Figure S1**


a
Global sensitivity of BRCA1-ONSET penetrance
$\mu_{TP53}$
$\theta_{D2}$
$\delta_{low}$
$\nu$
$\beta$
0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35
Relative nonlinear importance
b
Age (years)
30 40 50 60 70 80
$\beta$
$\mu_{TP53}$
$\delta_{low}$
$\delta_{high}/\delta_{low}$
$\theta_{D1}$
$\theta_{D2}$
$d$
$\kappa_{BRCA1}$
$a$
$\nu$
PRCC
0.8
0.0
−0.8
Posterior empirical cumulative distribution functions (ECDFs) versus uniform priors
c
$\beta$
$\delta_{low}$
$\delta_{high}$
$\nu$
$\mu_{TP53}$
$\theta_{D1}$
$\theta_{D2}$
$d$
$\kappa_{BRCA1}$
Uniform Prior
Global
Locked
CDF